



Top 45 CAT Time and Work Questions With Video Solutions

All rights reserved. No part of this publication may be reproduced, distributed, or transmitted in any form or by any means, electronic, mechanical, photocopying, recording or otherwise, or stored in any retrieval system of any nature without the permission of cracku.in, application for which shall be made to support@cracku.in

Questions

Instructions

For the following questions answer them individually

Question 1

A chemical plant has four tanks (A, B, C, and D), each containing 1000 litres of a chemical. The chemical is being pumped from one tank to another as follows:

From A to B @ 20 litres/minute

From C to A @ 90 litres/minute

From A to D @ 10 litres/minute

From C to D @ 50 litres/minute

From B to C @ 100 litres/minute

From D to B @ 110 litres/minute

Which tank gets emptied first, and how long does it take (in minutes) to get empty after pumping starts?

- A A,16.66
- B C,20
- C D,20
- D D,25

[VIEW SOLUTION](#)

Question 2

A can complete a piece of work in 4 days. B takes double the time taken by A, C takes double that of B, and D takes double that of C to complete the same task. They are paired in groups of two each. One pair takes two-thirds the time needed by the second pair to complete the work. Which is the first pair?

- A A and B
- B A and C
- C B and C
- D A and D

cracku

A can complete a piece of work in 4 days. B takes double the time taken by A, C takes double that of B, and D takes double that of C to complete the same task. They are paired in groups of two each. One pair takes two-thirds the time needed by the second pair to complete the work. Which is the first pair?

A) A and B
B) A and C
C) B and C
D) A and D

Handwritten solution:

(8) A $\rightarrow$ 4 days
(4) B $\rightarrow$ 8 days
(2) C $\rightarrow$ 16 days
(1) D $\rightarrow$ 32 days
15 units/day

Total work = 32 units

Diagram showing two pairs: (A, B) and (C, D) with arrows indicating work flow.

Question 3

There's a lot of work in preparing a birthday dinner. Even after the turkey is in the oven, there's still the potatoes and gravy, yams, salad, and cranberries, not to mention setting the table.

Three friends — Asit, Arnold and Afzal — work together to get all of these chores done. The time it takes them to do the work together is 6 hr less than Asit would have taken working alone, 1 hr less than Arnold would have taken alone, and half the time Afzal would have taken working alone. How long did it take them to do these chores working together?

- A 20 min
- B 30 min
- C 40 min
- D 50 min

Handwritten solution for Question 3:

Let the time taken by Asit, Arnold, and Afzal working alone be $A+6$, $A+1$, and A hours respectively.

The time taken by all three working together is A hours.

According to the problem:

$$\frac{1}{A} = \frac{1}{A+6} + \frac{1}{A+1} + \frac{1}{2A}$$

Solving this equation, we get $A = 30$ min.

Options: A) 20 min, B) 30 min, C) 40 min, D) 50 min.

CAT Mock Test

Mocks designed exactly like the actual CAT

Question 4

It takes six technicians a total of 10 hr to build a new server from Direct Computer, with each working at the same rate. If six technicians start to build the server at 11 am, and one technician per hour is added beginning at 5 pm, at what time will the server be completed?

[CAT 2002]

- A 6.40 pm
- B 7 pm
- C 7.20 pm
- D 8 pm

It takes six technicians a total of 10 hr to build a new server from Direct Computer, with each working at the same rate. If six technicians start to build the server at 11 am, and one technician per hour is added beginning at 5 pm, at what time will the server be completed?

[CAT 2002]

A) 6:40 pm
B) 7 pm
C) 7:20 pm
D) 8 pm

Handwritten notes:
 total work = 60 units
 6 - 10 hrs
 Tech → unit hour
 11 am - 5 pm
 6 hrs
 36
 6 tech

VIDEO SOLUTION

Question 5

Three small pumps and a large pump are filling a tank. Each of the three small pump works at $\frac{2}{3}$ the rate of the large pump. If all four pumps work at the same time, they should fill the tank in what fraction of the time that it would have taken the large pump alone?

- A $\frac{4}{7}$
- B $\frac{1}{3}$
- C $\frac{2}{3}$
- D $\frac{3}{4}$

Three small pumps and a large pump are filling a tank. Each of the three small pump works at $\frac{2}{3}$ the rate of the large pump. If all four pumps work at the same time, they should fill the tank in what fraction of the time that it would have taken the large pump alone?

[CAT 2002]

A) $\frac{4}{7}$
B) $\frac{1}{3}$
C) $\frac{2}{3}$
D) $\frac{3}{4}$

Handwritten notes:
 S S S
 2r
 L
 3r
 9r

VIDEO SOLUTION

Question 6

A, B and C individually can finish a work in 6, 8 and 15 hours respectively. They started the work together and after completing the work got Rs.94.60 in all. When they divide the money among themselves, A, B and C will respectively get (in Rs.)

- A 44, 33, 17.60
- B 43, 27, 20, 24.40
- C 45, 30, 19.60
- D 42, 28, 24.60

 VIDEO SOLUTION

Question 7

A 12,4

B 10,3

C 10,4

D None of these

 [VIEW SOLUTION](#)

A 12 hours

B 10 hours

C 8 hours

D 6 hour

Three machines A, B and C can be used to produce a product. Machine A will take 60 hours to produce a million units. Machine B is twice as fast as Machine A. Machine C will take the same amount of time to produce a million units as A and B running together. How much time will be required to produce a million units if all the three machines are used simultaneously?

A) 12 hours
B) 10 hours
C) 8 hours
D) 6 hours

Handwritten notes: 60 units, 1 hr → A → 1 unit, A - 60 hours, B - 30 hours.



VIDEO SOLUTION

Question 9

A company has a job to prepare certain number cans and there are three machines A, B and C for this job. A can complete the job in 3 days, B can complete the job in 4 days, and C can complete the job in 6 days. How many days will the company take to complete the job if all the machines are used simultaneously?

- A 4 days
- B $\frac{4}{3}$ days
- C 3 days
- D 12 days

A company has a job to prepare certain number cans and there are three machines A, B and C for this job. A can complete the job in 3 days, B can complete the job in 4 days, and C can complete the job in 6 days. How many days will the company take to complete the job if all the machines are used simultaneously?

A) 4 days
B) $\frac{4}{3}$ days
C) 3 days

Handwritten notes: Job = LCM(3, 4, 6), A → 3 days, B → 4 days, C → 6 days.



VIDEO SOLUTION

CAT Previous Papers

with detailed video solutions and analysis




Question 10

A person can complete a job in 120 days. He works alone on Day 1. On Day 2, he is joined by another person who also can complete the job in exactly 120 days. On Day 3, they are joined by another person of equal efficiency. Like this, everyday a new person with the same efficiency joins the work. How many days are required to complete the job?

cracku

A person can complete a job in 120 days. He works alone on Day 1. On Day 2, he is joined by another person who also can complete the job in exactly 120 days. On Day 3, they are joined by another person of equal efficiency. Like this, everyday a new person with the same efficiency joins the work. How many days are required to complete the job?

Total work = 120 units
 1 person = 1 unit/day $\rightarrow$ 120 days.

1 $\rightarrow$ 1 unit
 2 $\rightarrow$ 1 + 1 = 2 units
 3 $\rightarrow$ 1 + 1 + 1 = 3 units
 4 $\rightarrow$ 4
 5 $\rightarrow$ 5



VIDEO SOLUTION

Question 11

A tank has an inlet pipe and an outlet pipe. If the outlet pipe is closed then the inlet pipe fills the empty tank in 8 hours. If the outlet pipe is open then the inlet pipe fills the empty tank in 10 hours. If only the outlet pipe is open then in how many hours the full tank becomes half-full?

- A 20
- B 30
- C 40
- D 45

cracku

A tank has an inlet pipe and an outlet pipe. If the outlet pipe is closed then the inlet pipe fills the empty tank in 8 hours. If the outlet pipe is open then the inlet pipe fills the empty tank in 10 hours. If only the outlet pipe is open then in how many hours the full tank becomes half-full?

$\frac{1}{8} \rightarrow 8 \text{ hrs}$
 $\frac{1}{10} \rightarrow 10 \text{ hrs}$
 $\frac{1}{8} - \frac{1}{10} = \frac{10 - 8}{80} = \frac{2}{80} = \frac{1}{40}$
 $\frac{1}{40} = 40 \text{ hrs}$

$V = 80 \text{ litres}$
 $I = 10 \text{ litres/hr}$



VIDEO SOLUTION

Question 12

Amal can complete a job in 10 days and Bimal can complete it in 8 days. Amal, Bimal and Kamal together complete the job in 4 days and are paid a total amount of Rs 1000 as remuneration. If this amount is shared by them in proportion to their work, then Kamal's share, in rupees, is

- A 100
- B 200
- C 300
- D 400

cracku

Amal can complete a job in 10 days and Bimal can complete it in 8 days. Amal, Bimal and Kamal together complete the job in 4 days and are paid a total amount of Rs 1000 as remuneration. If this amount is shared by them in proportion to their work, then Kamal's share, in rupees, is

$\frac{4 \text{ units/day}}{5 \text{ units/day}} = A \rightarrow \frac{10 \text{ days}}{8 \text{ days}}$
 $\frac{4 \text{ units/day}}{5 \text{ units/day}} = B \rightarrow \frac{10 \text{ days}}{8 \text{ days}}$
 $A+B+K \rightarrow 4 \text{ day}$

Total work = 40 units
 $\text{LCM}(10, 8, 4)$



VIDEO SOLUTION



CAT Syllabus PDF



Question 13

Humans and robots can both perform a job but at different efficiencies. Fifteen humans and five robots working together take thirty days to finish the job, whereas five humans and fifteen robots working together take sixty days to finish it. How many days will fifteen humans working together (without any robot) take to finish it?

- A 45
- B 36
- C 32
- D 40

cracku

Humans and robots can both perform a job but at different efficiencies. Fifteen humans and five robots working together take thirty days to finish the job, whereas five humans and fifteen robots working together take sixty days to finish it. How many days will fifteen humans working together (without any robot) take to finish it?

$\text{Total work} = 60 \text{ units}$
 $15H + 5R \rightarrow 30 \text{ days}$
 $5H + 15R \rightarrow 60 \text{ days}$
 $15H \rightarrow ?$

$\frac{1}{30} \text{ day} \cdot 15H + 5R = \frac{1}{60} \text{ day} \cdot 5H + 15R$
 $\frac{1}{30} \text{ day} \cdot 15H = \frac{1}{60} \text{ day} \cdot 5H$



VIDEO SOLUTION

Question 14

When they work alone, B needs 25% more time to finish a job than A does. They two finish the job in 13 days in the following manner: A works alone till half the job is done, then A and B work together for four days, and finally B works alone to complete the remaining 5% of the job. In how many days can B alone finish the entire job?

- A 20

B 22

C 16

D 18

cracku

When they work alone, B needs 25% more time to finish a job than A does. They two finish the job in 13 days in the following manner: A works alone till half the job is done, then A and B work together for four days, and finally B works alone to complete the remaining 5% of the job. In how many days can B alone finish the entire job?

$A \rightarrow 50\%$
 $A+B \rightarrow 4 \text{ days}$
 $B \rightarrow 5\%$

$A = 1.25 \text{ units/day}$
 $B = 1 \text{ unit/day}$

50% 1 day 5%



VIDEO SOLUTION

Question 15

A tank is fitted with pipes, some filling it and the rest draining it. All filling pipes fill at the same rate, and all draining pipes drain at the same rate. The empty tank gets completely filled in 6 hours when 6 filling and 5 draining pipes are on, but this time becomes 60 hours when 5 filling and 6 draining pipes are on. In how many hours will the empty tank get completely filled when one draining and two filling pipes are on?

cracku

A tank is fitted with pipes, some filling it and the rest draining it. All filling pipes fill at the same rate, and all draining pipes drain at the same rate. The empty tank gets completely filled in 6 hours when 6 filling and 5 draining pipes are on, but this time becomes 60 hours when 5 filling and 6 draining pipes are on. In how many hours will the empty tank get completely filled when one draining and two filling pipes are on?

60 hours

6 Hours

60 hours

$\frac{6F}{F} = \frac{5D}{6D}$

$2F = 1D$



VIDEO SOLUTION

cracku

CAT Formula PDF

PDF



Question 16

A water tank has inlets of two types A and B. All inlets of type A when open, bring in water at the same rate. All inlets of type B, when open, bring in water at the same rate. The empty tank is completely filled in 30 minutes if 10 inlets of type A and 45 inlets of type B are open, and in 1 hour if 8 inlets of type A and 18 inlets of type B are open. In how many minutes will the empty tank get completely filled if 7 inlets of type A and 27 inlets of type B are open?

cracku

A water tank has inlets of two types A and B. All inlets of type A when open, bring in water at the same rate. All inlets of type B, when open, bring in water at the same rate. The empty tank is completely filled in 30 minutes if 10 inlets of type A and 45 inlets of type B are open, and in 1 hour if 8 inlets of type A and 18 inlets of type B are open. In how many minutes will the empty tank get completely filled if 7 inlets of type A and 27 inlets of type B are open?

Solution:

$$\begin{aligned} 30 \text{ mins} &\rightarrow 10A + 45B \\ 60 \text{ mins} &\rightarrow 8A + 18B \\ 10A + 45B &= 2(8A + 18B) \\ 10A + 45B &= 16A + 36B \\ 8A + 9B &= 0 \end{aligned}$$

$7A + 27B = ?$

VIDEO SOLUTION

Question 17

A tank is emptied everyday at a fixed time point. Immediately thereafter, either pump A or pump B or both start working until the tank is full. On Monday, A alone completed filling the tank at 8 pm. On Tuesday, B alone completed filling the tank at 6 pm. On Wednesday, A alone worked till 5 pm, and then B worked alone from 5 pm to 7 pm, to fill the tank. At what time was the tank filled on Thursday if both pumps were used simultaneously all along?

- A 4:48 pm
- B 4:12 pm
- C 4:24 pm
- D 4:36 pm

cracku

A tank is emptied everyday at a fixed time point. Immediately thereafter, either pump A or pump B or both start working until the tank is full. On Monday, A alone completed filling the tank at 8 pm. On Tuesday, B alone completed filling the tank at 6 pm. On Wednesday, A alone worked till 5 pm, and then B worked alone from 5 pm to 7 pm, to fill the tank. At what time was the tank filled on Thursday if both pumps were used simultaneously all along?

Solution:

Let the tank be emptied at x pm.

On Monday, A alone completed filling the tank at 8 pm. $A \rightarrow (8-x)$ hrs.

On Tuesday, B alone completed filling the tank at 6 pm. $B \rightarrow (6-x)$ hrs.

On Wednesday, A alone worked till 5 pm, and then B worked alone from 5 pm to 7 pm, to fill the tank. $A \rightarrow (5-x)$ hrs, $B \rightarrow (7-5)$ hrs.

On Thursday, both pumps were used simultaneously all along. $A \rightarrow (8-x)$ hrs, $B \rightarrow (6-x)$ hrs.

VIDEO SOLUTION

Question 18

Ramesh and Ganesh can together complete a work in 16 days. After seven days of working together, Ramesh got sick and his efficiency fell by 30%. As a result, they completed the work in 17 days instead of 16 days. If Ganesh had worked alone after Ramesh got sick, in how many days would he have completed the remaining work?

- A 14.5
- B 11

C 13.5

D 12

cracku

Ramesh and Ganesh can together complete a work in 16 days. After seven days of working together, Ramesh got sick and his efficiency fell by 30%. As a result, they completed the work in 17 days instead of 16 days. If Ganesh had worked alone after Ramesh got sick, in how many days would he have completed the remaining work?

The handwritten solution shows a timeline of 16 days. The first 7 days are labeled 'R+G' and the next 9 days are labeled 'G'. To the right, the equations are written: $(R+G) \times 16 = 1$, $7(R+G) + 10(0.7R+G) = 1$, $7R+7G+7R+10G = 1$, $14R+17G = 1$, and $16R+16G = 1$.

VIDEO SOLUTION

IIM Call Predictor
Accurate analysis of your profile

Question 19

Anil alone can do a job in 20 days while Sunil alone can do it in 40 days. Anil starts the job, and after 3 days, Sunil joins him. Again, after a few more days, Bimal joins them and they together finish the job. If Bimal has done 10% of the job, then in how many days was the job done?

A 12

B 13

C 15

D 14

cracku

Anil alone can do a job in 20 days while Sunil alone can do it in 40 days. Anil starts the job, and after 3 days, Sunil joins him. Again, after a few more days, Bimal joins them and they together finish the job. If Bimal has done 10% of the job, then in how many days was the job done?

The handwritten solution shows a timeline with 6 units of work. The first 3 days are labeled 'A' and the next 3 days are labeled 'S'. To the right, the calculations are written: $A - 20 \text{ days} = 2 \text{ units} \rightarrow \text{Total} = 40 \text{ units}$, $S - 40 \text{ days} = 1 \text{ unit/day}$, and $3 \text{ days} = 2 \times 3 = 6 \text{ units}$.

VIDEO SOLUTION

Question 20

At their usual efficiency levels, A and B together finish a task in 12 days. If A had worked half as efficiently as she usually does, and B had worked thrice as efficiently as he usually does, the task would have been completed in 9 days. How many days would A take to finish the task if she works alone at her usual efficiency?

- A 36
- B 24
- C 18
- D 12

cracku

At their usual efficiency levels, A and B together finish a task in 12 days. If A had worked half as efficiently as she usually does, and B had worked thrice as efficiently as he usually does, the task would have been completed in 9 days. How many days would A take to finish the task if she works alone at her usual efficiency?

$\rightarrow A + B = \frac{1}{12}$
 $\frac{A}{2} + 3B = \frac{1}{9}$

$A \rightarrow \frac{10}{10} \leftarrow \frac{1}{20}$

VIDEO SOLUTION

Question 21

Three men and eight machines can finish a job in half the time taken by three machines and eight men to finish the same job. If two machines can finish the job in 13 days, then how many men can finish the job in 13 days?

cracku

Three men and eight machines can finish a job in half the time taken by three machines and eight men to finish the same job. If two machines can finish the job in 13 days, then how many men can finish the job in 13 days?

$3a + 8b = 2 \times (3b + 8a) \leftarrow (1)$

$3a + 8b = 6b + 16a$
 $2b = 13a$

VIDEO SOLUTION

CAT Percentile Predictor



Question 22

A contractor agreed to construct a 6 km road in 200 days. He employed 140 persons for the work. After 60 days, he realized that only 1.5 km road has been completed. How many additional people would he need to employ in order to finish the work exactly on time?

A contractor agreed to construct a 6 km road in 200 days. He employed 140 persons for the work. After 60 days, he realized that only 1.5 km road has been completed. How many additional people would he need to employ in order to finish the work exactly on time?

$$\begin{array}{l}
 140 \rightarrow 60 \rightarrow 1.5 \text{ km} \\
 1 \rightarrow 1 \\
 140 \times 2 \\
 N \rightarrow 140
 \end{array}$$

VIDEO SOLUTION

Question 23

John takes twice as much time as Jack to finish a job. Jack and Jim together take one-thirds of the time to finish the job than John takes working alone. Moreover, in order to finish the job, John takes three days more than that taken by three of them working together. In how many days will Jim finish the job working alone?

cracku

John takes twice as much time as Jack to finish a job. Jack and Jim together take one-thirds of the time to finish the job than John takes working alone. Moreover, in order to finish the job, John takes three days more than that taken by three of them working together. In how many days will Jim finish the job working alone?

John — 1 unit/day
 Jack — 2 units/day
 Jack + Jim — 3 units/day
 Jack — 1 unit/day

Total work =

VIDEO SOLUTION

Question 24

Anu, Vinu and Manu can complete a work alone in 15 days, 12 days and 20 days, respectively. Vinu works everyday. Anu works only on alternate days starting from the first day while Manu works only on alternate days starting from the second day. Then, the number of days needed to complete the work is

- A 5
- B 8
- C 6
- D 7

Anu, Vinu and Manu can complete a work alone in 15 days, 12 days and 20 days, respectively. Vinu works everyday. Anu works only on alternate days starting from the first day while Manu works only on alternate days starting from the second day. Then, the number of days needed to complete the work is

A) 5
B) 6

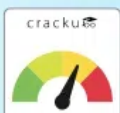
Handwritten notes:
Total work = LCM(15, 12, 20) = 60 units.
(4) A → 15 days
(5) V → 12 days
(6) M → 20 days

Handwritten list:
1) V (5)
2) V
3) V
4) V
5) V
6) V
7) V
8) V



VIDEO SOLUTION

CAT Score Calculator



Question 25

Amar, Akbar and Anthony are working on a project. Working together Amar and Akbar can complete the project in 1 year, Akbar and Anthony can complete in 16 months, Anthony and Amar can complete in 2 years. If the person who is neither the fastest nor the slowest works alone, the time in months he will take to complete the project is

Amar, Akbar and Anthony are working on a project. Working together Amar and Akbar can complete the project in 1 year, Akbar and Anthony can complete in 16 months, Anthony and Amar can complete in 2 years. If the person who is neither the fastest nor the slowest works alone, the time in months he will take to complete the project is

Handwritten notes:
Total work = 48 units
LCM(12, 16, 24) = 48
Amar - a
Akbar - b
Anthony - c
1 month
24 months



VIDEO SOLUTION

Question 26

Anil can paint a house in 60 days while Bimal can paint it in 84 days. Anil starts painting and after 10 days, Bimal and Charu join him. Together, they complete the painting in 14 more days. If they are paid a total of ₹ 21000 for the job, then the share of Charu, in INR, proportionate to the work done by him, is

- A 9000
- B 9200
- C 9100
- D 9150

cracku

Anil can paint a house in 60 days while Bimal can paint it in 84 days. Anil starts painting and after 10 days, Bimal and Charu join him. Together, they complete the painting in 14 more days. If they are paid a total of ₹ 21000 for the job, then the share of Charu, in INR, proportionate to the work done by him, is

A) 9000
B) 9200
C) 9100

Handwritten notes:
 $\frac{\text{7 units}}{\text{day}} \rightarrow A \rightarrow 60 \text{ days}$
 $\frac{\text{5 units}}{\text{day}} \rightarrow B \rightarrow 84 \text{ days}$
 $\frac{\text{units}}{\text{day}} \rightarrow C$
 $70 + 5 \times 14 + 14C$
 Total work = LCM (60, 84) = 420 units



VIDEO SOLUTION

Question 27

Two pipes A and B are attached to an empty water tank. Pipe A fills the tank while pipe B drains it. If pipe A is opened at 2 pm and pipe B is opened at 3 pm, then the tank becomes full at 10 pm. Instead, if pipe A is opened at 2 pm and pipe B is opened at 4 pm, then the tank becomes full at 6 pm. If pipe B is not opened at all, then the time, in minutes, taken to fill the tank is

- A 144
- B 140
- C 264
- D 120

cracku

Two pipes A and B are attached to an empty water tank. Pipe A fills the tank while pipe B drains it. If pipe A is opened at 2 pm and pipe B is opened at 3 pm, then the tank becomes full at 10 pm. Instead, if pipe A is opened at 2 pm and pipe B is opened at 4 pm, then the tank becomes full at 6 pm. If pipe B is not opened at all, then the time, in minutes, taken to fill the tank is

A) 144
B) 140
C) 264

Handwritten notes:
 $V = 8A - 7B$
 $V = 4A -$
 $\frac{A}{\text{hr}}$
 $\frac{B}{\text{hr}}$
 V
 $\rightarrow \text{units/hr}$



VIDEO SOLUTION

cracku

CAT Crash Course

Live Classes by IIMA alumni



Question 28

One day, Rahul started a work at 9 AM and Gautam joined him two hours later. They then worked together and completed the work at 5 PM the same day. If both had started at 9 AM and worked together, the work would have been completed 30 minutes earlier. Working alone, the time Rahul would have taken, in hours, to complete the work is

- A 11.5
- B 10
- C 12.5
- D 12

One day, Rahul started a work at 9 AM and Gautam joined him two hours later. They then worked together and completed the work at 5 PM the same day. If both had started at 9 AM and worked together, the work would have been completed 30 minutes earlier. Working alone, the time Rahul would have taken, in hours, to complete the work is

A) 11.5
B) 10
C) 12.5

Handwritten notes: $8R + 6G = W$, $R \rightarrow 9am$, $G \rightarrow 11am$, $(9am)$

VIDEO SOLUTION

Question 29

Anil can paint a house in 12 days while Barun can paint it in 16 days. Anil, Barun, and Chandu undertake to paint the house for ₹ 24000 and the three of them together complete the painting in 6 days. If Chandu is paid in proportion to the work done by him, then the amount in INR received by him is

Anil can paint a house in 12 days while Barun can paint it in 16 days. Anil, Barun, and Chandu undertake to paint the house for ₹ 24000 and the three of them together complete the painting in 6 days. If Chandu is paid in proportion to the work done by him, then the amount in INR received by him is

Answer: _____

Handwritten notes: 4 units/day , $A \rightarrow 12 \text{ days}$, $B \rightarrow 16 \text{ days}$, $A+B \rightarrow 6 \text{ days}$, $\text{Total} = LCM(12, 16, 5) = 48 \text{ units}$

VIDEO SOLUTION

Question 30

Working alone, the times taken by Anu, Tanu and Manu to complete any job are in the ratio 5 : 8 : 10. They accept a job which they can finish in 4 days if they all work together for 8 hours per day. However, Anu and Tanu work together for the first 6 days, working 6 hours 40 minutes per day. Then, the number of hours that Manu will take to complete the remaining job working alone is

Working alone, the times taken by Anu, Tanu and Manu to complete any job are in the ratio 5 : 8 : 10. They accept a job which they can finish in 4 days if they all work together for 8 hours per day. However, Anu and Tanu work together for the first 6 days, working 6 hours 40 minutes per day. Then, the number of hours that Manu will take to complete the remaining job working alone is

Answer: _____

32

Efficiency

A : T : M = 5 : 8 : 10

$\frac{1}{5} : \frac{1}{8} : \frac{1}{10}$

$\times 40$

8 : 5 : 4

units/hr

units/hr

units/hr

A = T

cracku

VIDEO SOLUTION

5000+ CAT Questions

"Cracku CAT Questions"

cracku

Question 31

A group of N people worked on a project. They finished 35% of the project by working 7 hours a day for 10 days. Thereafter, 10 people left the group and the remaining people finished the rest of the project in 14 days by working 10 hours a day. Then the value of N is

- A 150
- B 23
- C 36
- D 140

A group of N people worked on a project. They finished 35% of the project by working 7 hours a day for 10 days. Thereafter, 10 people left the group and the remaining people finished the rest of the project in 14 days by working 10 hours a day. Then the value of N is

A) 150

B) 23

C) 36

D) 140

Person $\rightarrow$ unit/hour

$7 \times 10 = 70$ hours

Total work done = 35%

Total work = 100%

$\frac{70}{100} = \frac{35}{100}$

$70 \times 10 = 35 \times 100$

$700 = 3500$

$N = 140$

cracku

VIDEO SOLUTION

Question 32

Bob can finish a job in 40 days, if he works alone. Alex is twice as fast as Bob and thrice as fast as Cole in the same job. Suppose Alex and Bob work together on the first day, Bob and Cole work together on the second day, Cole and Alex work together on the third day, and then, they continue the work by repeating this three - day roster, with Alex and Bob working together on the fourth day, and so on. Then, the total number of days Alex would have worked when the job gets finished, is

cracku

Bob can finish a job in 40 days, if he works alone. Alex is twice as fast as Bob and thrice as fast as Cole in the same job. Suppose Alex and Bob work together on the first day, Bob and Cole work together on the second day, Cole and Alex work together on the third day, and then, they continue the work by repeating this three-day roster, with Alex and Bob working together on the fourth day, and so on. Then, the total number of days Alex would have worked when the job gets finished, is

Answer: _____

$A+B$
 $B+C$
 $A+C$

$B \rightarrow 40$ days (2 units/day)
 $A \rightarrow 20$ days (6 units/day)
 $C \rightarrow 60$ days (2 units/day)

Total work = LCM(20, 40, 60) = 120 units



VIDEO SOLUTION

Question 33

The amount of job that Amal, Sunil and Kamal can individually do in a day, are in harmonic progression. Kamal takes twice as much time as Amal to do the same amount of job. If Amal and Sunil work for 4 days and 9 days, respectively, Kamal needs to work for 16 days to finish the remaining job. Then the number of days Sunil will take to finish the job working alone, is

VIDEO SOLUTION

cracku

CAT Daily Targets

Step by step, heading towards 99+ percentile




Question 34

Rahul, Rakshita and Gurmeet, working together, would have taken more than 7 days to finish a job. On the other hand, Rahul and Gurmeet, working together would have taken less than 15 days to finish the job. However, they all worked together for 6 days, followed by Rakshita, who worked alone for 3 more days to finish the job. If Rakshita had worked alone on the job then the number of days she would have taken to finish the job, cannot be

- A 20
- B 17
- C 16
- D 21

cracku

Rahul, Rakshita and Gurmeet, working together, would have taken more than 7 days to finish a job. On the other hand, Rahul and Gurmeet, working together would have taken less than 15 days to finish the job. However, they all worked together for 6 days, followed by Rakshita, who worked alone for 3 more days to finish the job. If Rakshita had worked alone on the job then the number of days she would have taken to finish the job, cannot be

A) 20
 C) 16

$\text{Total work} > 7(A+B+C)$
 $\text{Total work} < 15(A+B)$

$\text{Total work} = 6(A+B+C) + 3B$
 $(A+B) + 3B > 7(A+B)$



VIDEO SOLUTION

Question 35

Gautam and Suhani, working together, can finish a job in 20 days. If Gautam does only 60% of his usual work on a day, Suhani must do 150% of her usual work on that day to exactly make up for it. Then, the number of days required by the faster worker to complete the job working alone is

Gautam and Suhani, working together, can finish a job in 20 days. If Gautam does only 60% of his usual work on a day, Suhani must do 150% of her usual work on that day to exactly make up for it. Then, the number of days required by the faster worker to complete the job working alone is

Answer: _____

$G + S = 0$

1 day
Gautam = G
Suhani = S
work = $20(G + S)$
 $20(0.6G + 1.5S)$

0.6G
1.5S

VIDEO SOLUTION

Question 36

Pipes A and C are fill pipes while Pipe B is a drain pipe of a tank. Pipe B empties the full tank in one hour less than the time taken by Pipe A to fill the empty tank. When pipes A, B and C are turned on together, the empty tank is filled in two hours. If pipes B and C are turned on together when the tank is empty and Pipe B is turned off after one hour, then Pipe C takes another one hour and 15 minutes to fill the remaining tank. If Pipe A can fill the empty tank in less than five hours, then the time taken, in minutes, by Pipe C to fill the empty tank is

- A 90
- B 120
- C 75
- D 60

Pipes A and C are fill pipes while Pipe B is a drain pipe of a tank. Pipe B empties the full tank in one hour less than the time taken by Pipe A to fill the empty tank. When pipes A, B and C are turned on together, the empty tank is filled in two hours. If pipes B and C are turned on together when the tank is empty and Pipe B is turned off after one hour, then Pipe C takes another one hour and 15 minutes to fill the remaining tank. If Pipe A can fill the empty tank in less than five hours, then the time taken, in minutes, by Pipe C to fill the empty tank is

A) 90
C) 75
D) 60

$\frac{V}{T}$ A $\rightarrow$ This $\uparrow$
 $-\frac{V}{T-1}$ B $\rightarrow (T-1)h$ $\downarrow$
 $\frac{V}{2} + \frac{V}{T(T-1)}$ C $\rightarrow$

$\frac{V}{T} - \frac{V}{T-1} + C = \frac{V}{2}$
 $C = \frac{V}{2} + \frac{V}{T(T-1)}$

VIDEO SOLUTION

Question 37

Amal and Vimal together can complete a task in 150 days, while Vimal and Sunil together can complete the same task in 100 days. Amal starts working on the task and works for 75 days, then Vimal takes over and works for 135 days. Finally, Sunil takes over and completes the remaining task in 45 days. If Amal had started the task alone and worked on all days, Vimal had worked on every second day, and Sunil had worked on every third day, then the number of days required to complete the task would have been

Amal and Vimal together can complete a task in 150 days, while Vimal and Sunil together can complete the same task in 100 days. Amal starts working on the task and works for 75 days, then Vimal takes over and works for 135 days. Finally, Sunil takes over and completes the remaining task in 45 days. If Amal had started the task alone and worked on all days, Vimal had worked on every second day, and Sunil had worked on every third day, then the number of days required to complete the task would have been

Answer: _____

1 day 2 units A+V → 150 days
3 units V+S → 100 days

75A + 35V + 4S = 300 units
5A + 7V + 4S = 200
5A + V = 10

Total work = LCM(150, 100)
= 300 units



[VIDEO SOLUTION](#)

Question 38

Renu would take 15 days working 4 hours per day to complete a certain task whereas Seema would take 8 days working 5 hours per day to complete the same task. They decide to work together to complete this task. Seema agrees to work for double the number of hours per day as Renu, while Renu agrees to work for double the number of days as Seema. If Renu works 2 hours per day, then the number of days Seema will work, is

Renu would take 15 days working 4 hours per day to complete a certain task whereas Seema would take 8 days working 5 hours per day to complete the same task. They decide to work together to complete this task. Seema agrees to work for double the number of hours per day as Renu, while Renu agrees to work for double the number of days as Seema. If Renu works 2 hours per day, then the number of days Seema will work, is

Answer: _____

1 hour Renu → 60 hours
3 Seema → 16 hours

Renu: x hrs/day

Total work = LCM(60, 40)
= 120 units



[VIDEO SOLUTION](#)

Question 39

Sam can complete a job in 20 days when working alone. Mohit is twice as fast as Sam and thrice as fast as Ayna in the same job. They undertake a job with an arrangement where Sam and Mohit work together on the first day, Sam and Ayna on the second day, Mohit and Ayna on the third day, and this three-day pattern is repeated till the work gets completed. Then, the fraction of total work done by Sam is

- A $\frac{1}{20}$
- B $\frac{3}{10}$
- C $\frac{1}{5}$
- D $\frac{3}{20}$

Sam can complete a job in 20 days when working alone. Mohit is twice as fast as Sam and thrice as fast as Ayna in the same job. They undertake a job with an arrangement where Sam and Mohit work together on the first day, Sam and Ayna on the second day, Mohit and Ayna on the third day, and this three-day pattern is repeated till the work gets completed. Then, the fraction of total work done by Sam is

cracku

Handwritten notes on a whiteboard:

- Sam = 20 days
- Mohit = 10 days
- Ayna = 30 days
- Total Work = LCM(20, 10, 30) = 60 units
- Sam's work = 2 units
- Mohit's work = 6 units
- Ayna's work = 2 units
- Options: A) $\frac{1}{20}$, B) $\frac{3}{10}$, C) $\frac{1}{5}$, D) $\frac{3}{20}$
- Equations: $S+M = 9$, $S+A = 9$



VIDEO SOLUTION

cracku

CAT Study Material

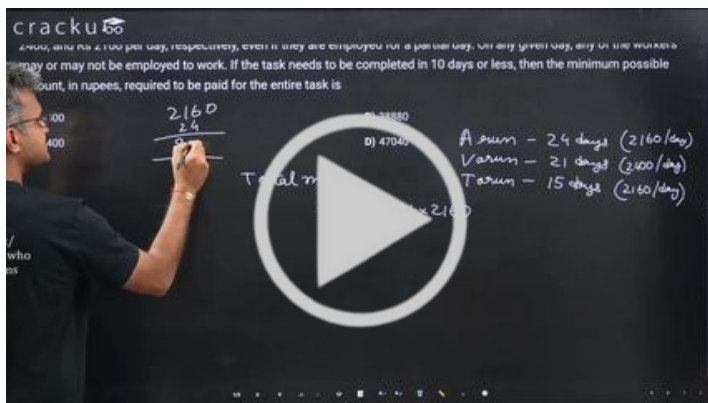
All CAT Free Resources



Question 40

Arun, Varun and Tarun, if working alone, can complete a task in 24, 21, and 15 days, respectively. They charge Rs 2160, Rs 2400, and Rs 2160 per day, respectively, even if they are employed for a partial day. On any given day, any of the workers may or may not be employed to work. If the task needs to be completed in 10 days or less, then the minimum possible amount, in rupees, required to be paid for the entire task is

- A 38400
- B 38880
- C 34400
- D 47040



VIDEO SOLUTION

Question 41

Teams A, B, and C consist of five, eight, and ten members, respectively, such that every member within a team is equally productive. Working separately, teams A, B, and C can complete a certain job in 40 hours, 50 hours, and 4 hours, respectively. Two members from team A, three members from team B, and one member from team C together start the job, and the member from team C leaves after 23 hours. The number of additional member(s) from team B, that would be required to replace the member from team C, to finish the job in the next one hour, is

- A 4
- B 2
- C 1
- D 3

VIEW SOLUTION

Question 42

The rate of water flow through three pipes A, B and C are in the ratio 4 : 9 : 36. An empty tank can be filled up completely by pipe A in 15 hours. If all the three pipes are used simultaneously to fill up this empty tank, the time, in minutes, required to fill up the entire tank completely is nearest to

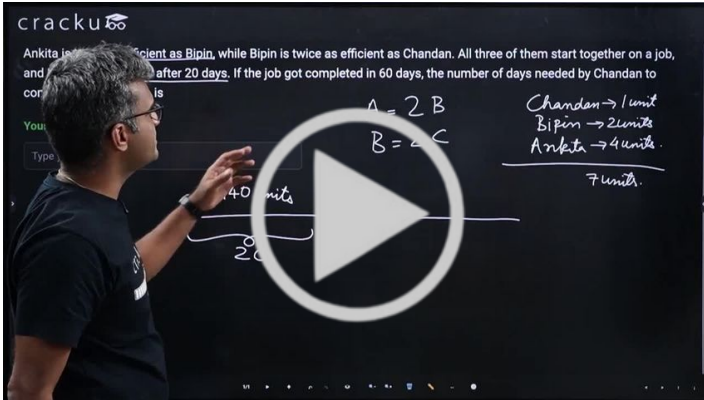
- A 73
- B 78
- C 76
- D 71

VIEW SOLUTION



Question 43

Ankita is twice as efficient as Bipin, while Bipin is twice as efficient as Chandan. All three of them start together on a job, and Bipin leaves the job after 20 days. If the job got completed in 60 days, the number of days needed by Chandan to complete the job alone, is



[VIDEO SOLUTION](#)

Instructions

The Weirdo Holiday Resort follows a particular system of holidays for its employees. People are given holidays on the days where the first letter of the day of the week is the same as the first letter of their names. All employees work at the same rate.

Question 44

Raja starts working on February 25(Sunday), 1996, and finishes the job on March 2, 1996. How much time would T and J take to finish the same job if both start on the same day as Raja?

- A 4 days
- B 5 days
- C Either a or b
- D Cannot be determined

[VIEW SOLUTION](#)

Question 45

Starting on February 25, 1996 (Sunday), if Raja had finished his job on April 2, 1996, when would T and S together likely to have completed the job, had they started on the same day as Raja?

- A March 15, 1996
- B March 14, 1996
- C March 22, 1996
- D Data insufficient

[VIEW SOLUTION](#)

100+ CAT Quant Questions



Answers

1.C	2.D	3.C	4.D	5.B	6.A	7.C	8.B
9.B	10.15	11.A	12.A	13.C	14.A	15.10	16.48
17.C	18.C	19.B	20.C	21.13	22.40	23.4	24.D
25.32	26.C	27.A	28.B	29.3000	30.6	31.D	32.11
33.27	34.D	35.36	36.A	37.139	38.6	39.B	40.A
41.B	42.A	43.340	44.A	45.C			

Explanations

1. C

After 1 min the cans will contain following amount of chemicals : A - 1060 B - 1030 C - 970 D - 950

So, we can see that the can D loses 50 ltrs in 1 min which is highest. So the can D will lose 1000 ltrs in $20 \times 1 = 20$ mins.

[VIEW SOLUTION](#)

2. D

A takes 4 days to complete the work.

So, B takes 8 days to complete the same work.

C takes 16 days to complete the work.

D takes 32 days to complete the same work.

In order to measure, let the total work be of 64 units. Hence, the speed of working of each of the four persons is given below.

A - 16 units/hr

B - 8 units/hr

C - 4 units/hr

D - 2 units/hr

From the given options, we need to find two pairs in such a way that their speeds are in the ratio 3:2. Note that $A+D=18$ while $B+C=12$ and the ratio is 3:2

Hence, the first pair is A and D and the second pair is B and C

[VIDEO SOLUTION](#)

3. C

Let the time taken working together be t.

Time taken by Arnold = $t+1$

Time taken by Asit = $t+6$

Time taken by Afzal = $2t$

Work done by each person in one day = $\frac{1}{(t+1)} + \frac{1}{(t+6)} + \frac{1}{2t}$

Total portion of workdone in one day = $\frac{1}{t}$

$$\frac{1}{(t+1)} + \frac{1}{(t+6)} + \frac{1}{2t} = \frac{1}{t}$$

$$\frac{1}{(t+1)} + \frac{1}{(t+6)} = \frac{2-1}{2t}$$

$$2t + 7 = \frac{(t+1) \cdot (t+6)}{2t}$$

$$3t^2 - 7t + 6 = 0 \rightarrow t = \frac{2}{3} \text{ or } t = -3$$

Therefore total time = $\frac{2}{3}$ hours = 40mins

Alternatively,

$$\frac{1}{(t+1)} + \frac{1}{(t+6)} + \frac{1}{2t} = \frac{1}{t}$$

From the options, if time = 40 min, that is, $t = \frac{2}{3}$

$$\text{LHS} = \frac{3}{5} + \frac{3}{20} + \frac{3}{4} = \frac{(12+3+15)}{20} = \frac{30}{20} = \frac{3}{2}$$

$$\text{RHS} = \frac{1}{t} = \frac{3}{2}$$

The equation is satisfied only in case of option C

Hence, C is correct

 VIDEO SOLUTION

4. D

Let the work done by each technician in one hour be 1 unit.

Therefore, total work to be done = 60 units.

From 11 AM to 5 PM, work done = $6 \times 6 = 36$ units.

Work remaining = $60 - 36 = 24$ units.

Work done in the next 3 hours = 7 units + 8 units + 9 units = 24 units.

Therefore, the work gets done by 8 PM.

 VIDEO SOLUTION

5. B

Let the work done by the big pump in one hour be 3 units.

Therefore, work done by each of the small pumps in one hour = 2 units.

Let the total work to be done in filling the tank be 9 units.

Therefore, time taken by the big pump if it operates alone = $9/3 = 3$ hours.

If all the pumps operate together, the work done in one hour = $3 + 2 \times 3 = 9$ units.

Together, all of them can fill the tank in 1 hour.

Required ratio = $1/3$

 VIDEO SOLUTION

100+ CAT DILR Questions



6. A

Money will be distributed in the ratio of work done in an hour.

$$\text{i.e. } \frac{1}{6} : \frac{1}{8} : \frac{1}{15}$$

or 20:15:8

$$\text{Hence part of A will be} = \frac{20}{43} \times 94.6 = 44$$

$$\text{part of B will be} = \frac{15}{43} \times 94.6 = 33$$

$$\text{part of C will be} = \frac{8}{43} \times 94.6 = 17.60$$

 VIDEO SOLUTION

7. C

Suppose he appointed x persons and y of them didn't come. Hence work done by each of them increases by 32.

$$\text{So } \frac{480}{x-y} - \frac{480}{x} = 32$$

Now we can check options by putting in the above eq.

x=10 and y=4 will be our answer

VIEW SOLUTION

8. B

As machine B's efficiency is twice as of A's, Hence, it will complete its work in 30 hours.

And C's efficiency is putting A and B together i.e. = 20 hours $((\frac{1}{60} + \frac{1}{30})^{-1})$

Now if all three work together, then it will be completed in x (say) days.

$$\frac{1}{x} = \frac{1}{20} + \frac{1}{30} + \frac{1}{60}$$

or x = 10 hours

VIDEO SOLUTION

9. B

If they work together, then total work done in single day = $\frac{1}{3} + \frac{1}{4} + \frac{1}{6} = \frac{9}{12}$

So $\frac{9}{12}$ work is done in 1 day

Hence unit work will be done in $\frac{12}{9}$ or $\frac{4}{3}$ days.

VIDEO SOLUTION

10. 15

Let the rate of work of a person be x units/day. Hence, the total work = 120x.

It is given that one first day, one person works, on the second day two people work and so on.

Hence, the work done on day 1, day 2,... will be x, 2x, 3x, ... respectively.

The sum should be equal to 120x.

$$120x = x * \frac{n(n+1)}{2}$$

$$n^2 + n - 240 = 0$$

n = 15 is the only positive solution.

Hence, it takes 15 days to complete the work.

VIDEO SOLUTION

100+ CAT VARC Questions



11. A

Let the time taken by the outlet pipe to empty = x hours

$$\text{Then, } \frac{1}{8} - \frac{1}{x} = \frac{1}{10}$$

$$\Rightarrow x = 40$$

Hence time taken by the outlet pipe to make the tank half-full = $40/2 = 20$ hour

VIDEO SOLUTION

12. A

Let the time take by kamal to complete the task be x days.

$$\text{Hence we have } \frac{1}{10} + \frac{1}{8} + \frac{1}{x} = \frac{1}{4}$$

$$\Rightarrow x = 40 \text{ days.}$$

Ratio of the work done by them = $\frac{1}{10} : \frac{1}{8} : \frac{1}{40} = 4 : 5 : 1$

Hence the wage earned by Kamal = $\frac{1}{10} * 1000 = 100$

 VIDEO SOLUTION

13. C

Let the efficiency of humans be 'h' and the efficiency of robots be 'r'.

In the first case,

Total work = $(15h + 5r) * 30$(i)

In the second case,

Total work = $(5h + 15r) * 60$(ii)

On equating (i) and (ii), we get

$(15h + 5r) * 30 = (5h + 15r) * 60$

Or, $15h + 5r = 10h + 30r$

Or, $5h = 25r$

Or, $h = 5r$

Total work = $(15h + 5r) * 30 = (15h + h) * 30 = 480h$

Time taken by 15 humans = $\frac{480h}{15h}$ days = 32 days.

Hence, option C is the correct answer.

 VIDEO SOLUTION

14. A

Let us assume that A can complete 'a' units of work in a day and B can complete 'b' units of work in a day.

A works alone till half the work is completed.

A and B work together for 4 days and B works alone to complete the last 5% of the work.

=> A and B in 4 days can complete 45% of the work.

Let us assume the total amount of work to be done to be 100 units.

$4a + 4b = 45$ -----(1)

B needs 25% more time than A to finish a job.

=> $1.25*b = a$ -----(2)

Substituting (2) in (1), we get,

$5b + 4b = 45$

$9b = 45$

$b = 5$ units/day

B alone can finish the job in $100/5 = 20$ days.

Therefore, option A is the right answer.

 VIDEO SOLUTION

15. 10

Let the efficiency of filling pipes be 'x' and the efficiency of draining pipes be 'y'.

In the first case,

Capacity of tank = $(6x - 5y) * 6$(i)

In the second case,

Capacity of tank = $(5x - 6y) * 60$(ii)

On equating (i) and (ii), we get

$(6x - 5y) * 6 = (5x - 6y) * 60$

or, $6x - 5y = 50x - 60y$

$$\text{or, } 44x = 55y$$

$$\text{or, } 4x = 5y$$

$$\text{or, } x = 1.25y$$

$$\text{Capacity of the tank} = (6x - 5y) * 6 = (7.5y - 5y) * 6 = 15y$$

$$\text{Net efficiency of 2 filling and 1 draining pipes} = (2x - y) = (2.5y - y) = 1.5y$$

$$\text{Time required} = \frac{15y}{1.5y} \text{ hours} = 10 \text{ hours.}$$

Hence, 10 is the correct answer.

 VIDEO SOLUTION

CAT Expected Questions PDF

cracku



16. 48

Let the efficiency of type A pipe be 'a' and the efficiency of type B be 'b'.

In the first case, 10 type A and 45 type B pipes fill the tank in 30 mins.

$$\text{So, the capacity of the tank} = \frac{1}{2}(10a + 45b) \dots (i)$$

In the second case, 8 type A and 18 type B pipes fill the tank in 1 hour.

$$\text{So, the capacity of the tank} = (8a + 18b) \dots (ii)$$

Equating (i) and (ii), we get

$$10a + 45b = 16a + 36b$$

$$\Rightarrow 6a = 9b$$

$$\text{From (ii), capacity of the tank} = (8a + 18b) = (8a + 12a) = 20a$$

In the third case, 7 type A and 27 type B pipes fill the tank.

$$\text{Net efficiency} = (7a + 27b) = (7a + 18a) = 25a$$

$$\text{Time taken} = \frac{20a}{25a} \text{ hour} = 48 \text{ minutes.}$$

Hence, 48 is the correct answer.

 VIDEO SOLUTION

17. C

Let 't' pm be the time when the tank is emptied everyday. Let 'a' and 'b' be the liters/hr filled by pump A and pump B respectively.

On Monday, A alone completed filling the tank at 8 pm. Therefore, we can say that pump A worked for (8 - t) hours. Hence, the volume of the tank = a*(8 - t) liters.

Similarly, on Tuesday, B alone completed filling the tank at 6 pm. Therefore, we can say that pump B worked for (6 - t) hours. Hence, the volume of the tank = b*(6 - t) liters.

On Wednesday, A alone worked till 5 pm, and then B worked alone from 5 pm to 7 pm, to fill the tank. Therefore, we can say that pump A worked for (5 - t) hours and pump B worked for 2 hours. Hence, the volume of the tank = a*(5 - t) + 2b liters.

$$\text{We can say that } a*(8 - t) = b*(6 - t) = a*(5 - t) + 2b$$

$$a*(8 - t) = a*(5 - t) + 2b$$

$$\Rightarrow 3a = 2b \dots (1)$$

$$a*(8 - t) = b*(6 - t)$$

Using equation (1), we can say that

$$a * (8 - t) = \frac{3a}{2} * (6 - t)$$

$$t = 2$$

Therefore, we can say that the tank gets emptied at 2 pm daily. We can see that A takes 6 hours and pump B takes 4 hours alone.

Hence, working together both can fill the tank in $= \frac{6 \times 4}{6+4} = 2.4$ hours or 2 hours and 24 minutes.

The pumps started filling the tank at 2:00 pm. Hence, the tank will be filled by 4:24 pm.

 VIDEO SOLUTION

18. C

Let 'R' and 'G' be the amount of work that Ramesh and Ganesh can complete in a day.

It is given that they can together complete a work in 16 days. Hence, total amount of work = $16(R+G)$... (1)

For first 7 days both of them worked together. From 8th day, Ramesh worked at 70% of his original efficiency whereas Ganesh worked at his original efficiency. It took them 17 days to finish the same work. i.e. Ramesh worked at 70% of his original efficiency for 10 days.

$$\Rightarrow 16(R+G) = 7(R+G) + 10(0.7R+G)$$

$$\Rightarrow 16(R+G) = 14R + 17G$$

$$\Rightarrow R = 0.5G \dots (2)$$

$$\text{Total amount of work left when Ramesh got sick} = 16(R+G) - 7(R+G) = 9(R+G) = 9(0.5G+G) = 13.5G$$

$$\text{Therefore, time taken by Ganesh to complete the remaining work} = \frac{13.5G}{G} = 13.5 \text{ days.}$$

 VIDEO SOLUTION

19. B

Let the total work be LCM of 20, 40 = 40 units

Efficiency of Anil and Sunil is 2 units and 1 unit per day respectively.

Anil works alone for 3 days, so Anil must have completed 6 units.

Bimal completes 10% of the work while working along with Anil and Sunil.

Bimal must have completed 4 units.

The remaining 30 units of work is done by Anil and Sunil

Number of days taken by them $30/3=10$

The total work is completed in $3+10=13$ days

 VIDEO SOLUTION

20. C

Assuming A completes a units of work in a day and B completes B units of work in a day and the total work = 1 unit

$$\text{Hence, } 12(a+b)=1 \dots (1)$$

$$\text{Also, } 9\left(\frac{a}{2}+3b\right)=1 \dots (2)$$

$$\text{Using both equations, we get, } 12(a+b)=9\left(\frac{a}{2}+3b\right)$$

$$\Rightarrow 4a+4b=\frac{3a}{2}+9b$$

$$\Rightarrow \frac{5a}{2}=5b$$

$$\Rightarrow a=2b$$

Substituting the value of b in equation (1),

$$12\left(\frac{3a}{2}\right)=1$$

$$\Rightarrow a = \frac{1}{18}$$

Hence, the number of days required = $1/\left(\frac{1}{18}\right)=18$

 VIDEO SOLUTION



21.13

Consider the work done by a man in a day = a and that by a machine = b

Since, three men and eight machines can finish a job in half the time taken by three machines and eight men to finish the same job, hence the efficiency will be double.

$$\Rightarrow 3a+8b = 2(3b+8a)$$

$$\Rightarrow 13a=2b$$

Hence work done by 13 men in a day = work done by 2 machines in a day.

$\Rightarrow$ If two machines can finish the job in 13 days, then same work will be done 13 men in 13 days.

Hence the required number of men = 13

 VIDEO SOLUTION

22.40

Let the desired efficiency of each worker '6x' per day.

$$140 \cdot 6x \cdot 200 = 6 \text{ km} \dots (i)$$

In 60 days $60/200 \cdot 6 = 1.8$ km of work is to be done but actually 1.5km is only done.

Actual efficiency 'y' = $1.5/1.8 \cdot 6x = 5x$.

Now, left over work = 4.5km which is to be done in 140 days with 'n' workers whose efficiency is 'y'.

$$\Rightarrow n \cdot 5x \cdot 140 = 4.5 \dots (ii)$$

(i)/(ii) gives,

$$\frac{(140 \cdot 6x \cdot 200)}{(n \cdot 5x \cdot 140)} = \frac{6}{4.5}$$

$$\Rightarrow n = 180.$$

$\therefore$ Extra $180-140 = 40$ workers are needed.

 VIDEO SOLUTION

23.4

Let Jack take "t" days to complete the work, then John will take "2t" days to complete the work. So work done by Jack in one day is $(1/t)$ and John is $(1/2t)$.

Now let Jim take "m" days to complete the work. According to question, $\frac{1}{t} + \frac{1}{m} = \frac{3}{2t}$ or $\frac{1}{m} = \frac{1}{2t}$ or $m = 2t$

Hence Jim takes "2t" time to complete the work.

Now let the three of them complete the work in "p" days. Hence John takes "p+3" days to complete the work.

$$\frac{1}{2t}(m+3) = \left(\frac{4}{2t}\right)m$$

$$\frac{1}{2t}(m+3) = \left(\frac{4}{2t}\right)m$$

or $m=1$. Hence Jim will take $(1+3)=4$ days to complete the work. Similarly John will also take 4 days to complete the work

 VIDEO SOLUTION

24. **D**

Let the total amount of work be 60 units.

Then Anu, Vinu, and Manu do 4, 5, and 3 units of work per day respectively.

On the 1st day, Anu and Vinu work. Work done on the 1st day = 9 units

On the 2nd day, Manu and Vinu work. Work done on the 2nd day = 8 units

This cycle goes on. And in 6 days, the work completed is $9+8+9+8+9+8 = 51$ units.

On the 7th day, again Anu and Vinu work and complete the remaining 9 units of work. Thus, the number of days taken is 7 days.

 VIDEO SOLUTION

25. **32**

Let the total work be 48 units. Let Amar do 'm' work, Akbar do 'k' work, and Anthony do 'n' units of work in a month.

Amar and Akbar complete the project in 12 months. Hence, in a month they do $\frac{48}{12}=4$ units of work.

$m+k = 4$.

Similarly, $k+n = 3$, and $m+n = 2$.

Solving the three equations, we get $m = \frac{3}{2}$, $k = \frac{5}{2}$, $n = \frac{1}{2}$.

Here, Amar works neither the fastest nor the slowest, and he does 1.5 units of work in a month. Hence, to complete the work, he would take $\frac{48}{1.5} = 32$ months.

 VIDEO SOLUTION



CAT WhatsApp Group

cracku



26. **C**

Let Entire work be W

Now Anil worked for 24 days

Bimal worked for 14 days and Charu worked for 14 days .

Now Anil Completes W in 60 days

so in 24 days he completed $0.4W$

Bimal completes W in 84 Days

So in 14 Days Bimal completes $= \frac{W}{6}$

Therefore work done by charu $= W - \frac{W}{6} - \frac{4W}{10} = \frac{26W}{10} = \frac{13W}{30}$

Therefore proportion of Charu $= \frac{13}{30} \times 21000 = 9100$

 VIDEO SOLUTION

27. **A**

Let A fill the tank at x liters/hour and B drain it at y liters/hour

Now as per Condition 1 :

We get Volume filled till 10pm $= 8x-7y$ (1) .

Here A operates for 8 hours and B operates for 7 hours .

As per condition 2

We get Volume filled till 6pm = $4x - 2y$ (2)

Here A operates for 4 hours and B operates for 2 hours .

Now equating (1) and (2)

we get $8x - 7y = 4x - 2y$

so we get $4x = 5y$

$y = 4x/5$

So volume of tank = $8x - 7 \times \frac{4x}{5} = \frac{12x}{5}$

So time taken by A alone to fill the tank = $\frac{\frac{12x}{5}}{x} = \frac{12}{5} \text{ hrs}$
= 144 minutes

 VIDEO SOLUTION

28. B

Let Rahul work at a units/hr and Gautam at b units/hour

Now as per the condition :

$8a + 6b = 7.5a + 7.5b$

so we get $0.5a = 1.5b$

or $a = 3b$

Therefore total work = $8a + 6b = 8a + 2a = 10a$

Now Rahul alone takes $10a/10 = 10$ hours.

 VIDEO SOLUTION

29. 3000

Now Anil Paints in 12 Days

Barun paints in 16 Days

Now together Arun , Barun and Chandu painted in 6 Days

Now let total work be W

Now each worked for 6 days

So Anil's work = $0.5W$

Barun's work = $\frac{6W}{16} = \frac{3W}{8}$

Therefore Charu's work = $\frac{W}{2} - \frac{3W}{8} = \frac{W}{8}$

Therefore proportion of charu = $\frac{24000}{8} = 3,000$

 VIDEO SOLUTION

30. 6

Let the time taken by Anu, Tanu and Manu be $5x$, $8x$ and $10x$ hours.

Total work = $\text{LCM}(5x, 8x, 10x) = 40x$

Anu can complete 8 units in one hour

Tanu can complete 5 units in one hour

Manu can complete 4 units in one hour

It is given, three of them together can complete in 32 hours.

$32(8 + 5 + 4) = 40x$

$x = \frac{68}{5}$

It is given,

Anu and Tanu work together for the first 6 days, working 6 hours 40 minutes per day, i.e. $36 + 4 = 40$ hours

$40(8 + 5) + y(4) = 40x$

$4y = 24$

$$y = 6$$

Manu alone will complete the remaining work in 6 hours.

 VIDEO SOLUTION



31. D

Let the unit of work done by 1 man in 1 hour and 1 day be 1 MDH unit (Man Day Hour).

Thus, in 7 hours per day for 10 days, the work done by N people = $N \times 7 \times 10$ MDH units.

Since this is equal to 35% of the total work,

35% of the total work = $N \times 7 \times 10$ MDH units.

Total work = $\frac{(N \times 100 \times 7 \times 10)}{35} = 200 \times N$ MDH units.

The work left = $200N - 70N = 130N$ MDH units.

Now, 10 people left the job. So, the number of people left = $(N-10)$

Since $(N-10)$ people completed the rest of work in 14 days by working 10 hours a day,

$$(N - 10) \times 14 \times 10 = 130N$$

$$10N = 1400$$

$$N = 140$$

Thus, the correct option is D.

 VIDEO SOLUTION

32. 11

Let the efficiency of Bob be 3 units/day. So, Alex's efficiency will be 6 units/day, and Cole's will be 2 units/day.

Since Bob can finish the job in 40 days, the total work will be $40 \times 3 = 120$ units.

Since Alex and Bob work on the first day, the total work done = $3 + 6 = 9$ units.

Similarly, for days 2 and 3, it will be 5 and 8 units, respectively.

Thus, in the first 3 days, the total work done = $9 + 5 + 8 = 22$ units.

The work done in the first 15 days = $22 \times 5 = 110$ units.

Thus, the work will be finished on the 17th day (since $9 + 5 = 14$ units are greater than the remaining work).

Since Alex works on two days of every 3 days, he will work for 10 days out of the first 15 days.

Then he will also work on the 16th day.

The total number of days = 11.

 VIDEO SOLUTION

33. 27

Let us assume the efficiencies of Amal, Sunil, and Kamal are a , s , and k , respectively.

Given that they are in H.P.

$$\Rightarrow \frac{2}{s} = \frac{1}{a} + \frac{1}{k} \text{ ---(1)}$$

Also, given that Kamal takes twice as much time as Amal to do the same amount of job

$$\Rightarrow a = 2k$$

Given that when Amal and Sunil work for 4 days and 9 days, respectively, Kamal needs to work for 16 days to finish the remaining job.

$$\Rightarrow \text{If } W \text{ is the total work } \Rightarrow 4a + 9s + 16k = W.$$

$$\text{from (1)} \frac{2}{s} = \frac{1}{a} + \frac{2}{a} \Rightarrow a = \frac{3}{2}s \text{ and } k = \frac{3}{4}s$$

$$\Rightarrow 4 \left(\frac{3s}{2} \right) + 9s + 16 \left(\frac{3s}{4} \right) = W$$

$$\Rightarrow 6s + 9s + 12s = W$$

$$\Rightarrow 27s = W \Rightarrow s = \frac{W}{27}$$

$\Rightarrow$ Sunil will take 27 days to finish the work when working alone.

 VIDEO SOLUTION

34. D

Let the work done by Rahul, Rakshita, and Gurmeet be a , b , and c units per day, respectively, and the total units of work are W .

Hence, we can say that $7(a+b+c) < W$ (Rahul, Rakshita, and Gurmeet, working together, would have taken more than 7 days to finish a job).

Similarly, we can say that $15(a+c) > W$ (Rahul and Gurmeet, working together would have taken less than 15 days to finish the job)

Now, comparing these two inequalities, we get: $7(a+b+c) < W < 15(a+c)$

It is also known that they all worked together for 6 days, followed by Rakshita, who worked alone for 3 more days to finish the job. Therefore, the total units of work done is: $W = 6(a+b+c) + 3b$

Hence, we can say that $7(a+b+c) < 6(a+b+c) + 3b < 15(a+c)$

Therefore, $(a+b+c) < 3b \Rightarrow a+c < 2b$, and $9b < 9(a+c) \Rightarrow b < a+c$

$\Rightarrow a+b+c < 3b \Rightarrow 7(a+b+c) < 21b$, and $15b < 15(a+c)$

Hence, The number of days required for b must be in between 15 and 21 (both exclusive).

Hence, the correct option is D

 VIDEO SOLUTION

35. 36

Let 'g' and 's' be the efficiencies of Gautam and Suhani. Let W is the total amount of work.

$$\Rightarrow g + s = W/20 \text{ (1 day work)} \text{ ----(1)}$$

Also Gautam doing only 60% $\Rightarrow 3g/5$ and Suhani doing 150% $\Rightarrow 3s/2$

$$\Rightarrow 3g/5 + 3s/2 = W/20 \text{ (1 day work)}$$

$$\Rightarrow g + s = \frac{3g}{5} + \frac{3s}{2}$$

$$\Rightarrow \frac{s}{g} = \frac{4}{5} \Rightarrow \text{Gautam is the more efficient person.}$$

Now, from the 1st equation

$$\Rightarrow g + \frac{4g}{5} = \frac{W}{20}$$

$$\Rightarrow \frac{9}{5}g = \frac{W}{20}$$

$$\Rightarrow g = \frac{W}{36}$$

$\Rightarrow$ Gautam takes 36 days to finish the complete work.

 VIDEO SOLUTION



36. A

Let the time taken by A to fill the tank alone be x hours, which implies the time taken by B to empty the tank alone is $(x-1)$ hours (B is the drainage pipe), and the time taken by C to fill the tank is y hours.

It is given that when pipes A, B, and C are turned on together, the empty tank is filled in two hours.

$$\text{Hence, } \frac{1}{x} - \frac{1}{x-1} + \frac{1}{y} = \frac{1}{2} \dots \text{Eq(1)}$$

It is given that if pipes B and C are turned on together when the tank is empty and Pipe B is turned off after one hour, then Pipe C takes another one hour and 15 minutes to fill the remaining tank.

Hence, B worked for 1 hour, and C worked for 2 hours 15 minutes, which is equal to $\frac{9}{4}$ hours.

In 1 hour, B worked $-\frac{1}{x-1}$ units, and in $\frac{9}{4}$ hours, C worked $\frac{9}{4y}$ units.

$$\text{Hence, } \frac{9}{4y} - \frac{1}{x-1} = 1 \dots \text{Eq(2)}$$

Solving both equations, we get $y = \frac{3}{2}$, and $x = 3$

Hence, the time taken by C is $\frac{3}{2}$ hours, which is equal to 90 minutes.

The correct option is A

 VIDEO SOLUTION

37. 139

We can take the work done by Amal, Vimal and Sunil to be A, V and S, respectively.

Let's take the total work they did to be T.

We are given the equations:

$$150A + 150V = T \dots (1)$$

$$100V + 100S = T \dots (2)$$

$$75A + 135V + 45S = T \dots (3)$$

$$\text{Adding (1) and (2), we get: } 150A + 250V + 100S = 2T \dots (4)$$

$$\text{And multiplying (3) with 2 we get: } 150A + 270V + 100S = 2T \dots (5)$$

Subtracting (5) from (4), we get $10S = 20V$ or simply $S = 2V$

Using this in (2), we get the total work T to be $300V$, and using that result in (1), we get $A=V$

Therefore, the work done by A, V and S equals V, V, and $2V$ units per day.

Now, in the question, we are given work cycles. To simplify the calculations, we should consider a time duration that is the LCM of the period taken by the three agents, which in this case would be 6

In 6 days, Amal will work for 6 days, doing $6V$ units of work.

Vimal will work for 3 days, doing $3V$ units of work.

Sunil will work for 2 days, doing $4V$ units of work.

So, in one 6-day cycle, $13V$ units of work will be done.

Dividing the total work ($300V$) by $13V$ we can see that in 23 cycles $299V$ units of work will be done.

These 23 cycles will be $23 \times 6 = 138$ days
The remaining 1V units of work will be done the next day.
Therefore, a total of 139 days will be required.

 VIDEO SOLUTION

38.6

Let us assign the amount of work done by Renu in one hour is R
And the amount of work done by Seema in one hour is S

We are told that a certain task with different time durations for Seema and Renu
Renu=15 days with 4 hours each day, that is total work done by Renu is $15 \times 4 \times R = 60R$
Similarly for Seema, 8 days and 5 hours each day, total work done by Seema is $40S$
We know that $60R = 40S$ or $S = 1.5R$

For this task, let us assume that the amount of days that Seema decides to work is X and the hours per day that Renu decides to work is Y

We are then told that, Seema works for 2Y hours per day and Renu works for 2X days,
Work done by them will be $2XYR$ and $2XY S$

We are told that $Y=2$, Making this $4XR$ and $4XS$
 $S = 1.5R$

Total work will be, $4XR+6XR=60R$

We get the value of $X = 6$

X is the number of days Seema will work, which is 6.

 VIDEO SOLUTION

39. B

We are given that Sam completes a piece of work in 20 days. We are also given that Mohit is twice as fast, so he should take only 10 days. Mohit is thrice as fast as Ayna, so he would take 30 days.

Let's take the total work to be 60 units; this would give the work done per day for Mohit, Sam, and Ayna to be 6, 3 and 2, respectively.

On the first day Sam and Mohit work: doing 9 units
On the second day Sam and Ayan work: doing 5 units
On the third day, Mohit and Ayan work: doing 8 units

Essentially doing 22 units in a 3 days cycle.
After two such cycles, there will be $60-44 = 16$ units of work left

On day 7, Sam and Mohit would work 9 units, leaving 7 units
On day 8, Sam and Ayan would work 5 units, leaving 2 units
And on day 9, Ayan and Mohit would complete the remaining work.

So Sam worked for a total of $2+2+2=6$ days and on each day he did 3 units of work, completing 18 units of work.

The ratio of work done by Sam would be $\frac{18}{60} = \frac{3}{10}$

Therefore, Option B is the correct answer.

 VIDEO SOLUTION

40. A

Let's assume that Total work = 1

If the work is entirely carried by each one of them it would cost

Arun: $2160 \times 24 = 51840$

Varun: $2400 \times 21 = 50400$

Tarun: $2160 \times 15 = 32400$

The cheapest worker is Tarun. He can finish the work in 15 days. But we need the work completed in 10 days.

Tarun does $10 \times \frac{1}{15} = \frac{2}{3}$

$1 - \frac{2}{3} = \frac{1}{3}$

Complete the remaining ($\frac{1}{3}$) work using the next cheapest worker, which is Varun.

Days needed: $\frac{x}{21} = \frac{1}{3} \Rightarrow x = 7$

Total cost = Amount paid for Tarun + Amount paid for Varun = $10 \times 2160 + 7 \times 2400 = 38400$

 VIDEO SOLUTION



41. B

Let the per hour productivity of each member from teams A, B, and C, be a , b , and c units respectively.

Let W be the total units of work that each of the teams does. We have:

$W = 40 * 5 * a = 50 * 8 * b = 4 * 10 * c$

This gives, $a = \frac{W}{200}$, $b = \frac{W}{400}$, and $c = \frac{W}{40}$

When two members from team A, three members from team B, and one member from team C together start the job, the total per hour productivity for the first 23 hours will be;

$\frac{2W}{200} + \frac{3W}{400} + \frac{W}{40} = \frac{17W}{400}$

In 23 hours, the work done will be $\frac{23 * 17W}{400} = \frac{391W}{400}$

The remaining work, after 23 hours, is $W - \frac{391W}{400} = \frac{9}{400}$

The current per hour efficiency of the group that comprises members from only teams A and B, is $\frac{2W}{200} + \frac{3W}{400} = \frac{7W}{400}$

In the next hour, the members from A and B will finish $\frac{7W}{400}$ units of work and $\frac{9W}{400} - \frac{7W}{400} = \frac{2W}{400}$ units of work would remain.

This work has to be finished by newly added members from team B in one hour, therefore, the number of new members required from team B would be $\frac{2W}{400} \div \frac{W}{400} = 2$

Option B is the correct answer.

 VIEW SOLUTION

42. A

Let the rate of flow of water from pipe A be $4x$ units per hour. From the ratio of the rates of flow of water (4:9:36), we have that the rate of flow of water from pipe B should be $9x$ units per hour, and from pipe C should be $36x$ units per hour.

Since pipe A fills the empty tank in 15 hours, the capacity of the tank must be $15 \times 4x = 60x$ units.

When all three pipes work together, their combined capacity would be $4x + 9x + 36x = 49x$ units per hour.

Therefore, the time it would take for the three pipes to fill the empty tank working together is: $\frac{60x}{49x}$ hours

In minutes, this time is equivalent to $\frac{60x}{49x} \times 60 \approx 73.47$ minutes.

Option A is the closest, and is the correct answer.

 [VIEW SOLUTION](#)

43. 340

Let Chandan's one-day work = x .

Then Bipin = $2x$ and Ankita = $4x$.

All three work together for the first 20 days. Their combined daily work = $x + 2x + 4x = 7x$.

Work done in 20 days = $20 \times 7x = 140x$.

Bipin leaves after 20 days; Ankita and Chandan continue for the remaining (60-20=40) days.

Their combined daily work = $4x + x = 5x$. Work done in 40 days = $40 \times 5x = 200x$.

Total work = $140x + 200x = 340x$.

We know that Chandan does x work in a day. So, the number of days it takes him to finish the work is 340.

Answer: 340 days.

 [VIDEO SOLUTION](#)

44. A

Raja takes 7 days to complete the job.

Feb 25, 1996 is a Sunday.

If T and J start working on Sunday, they can complete the work by Wednesday because T would have worked for 3 days and J would have worked for 4 days, thereby matching the number of days worked by Raja.

Hence, they can complete the job in 4 days.

 [VIEW SOLUTION](#)

45. C

The number of days taken by Raja to complete the work is $5+31+2 = 38$

So, cumulative number of days needed by T and S to complete the work is 38.

Both of them take two days off in a week as S takes off on Saturday and Sunday and T takes off on Tuesday and Thursday.

So, total number of man-working days per week by the duo is 10.

Hence, after three weeks, they finish 30 man working days.

i.e. by end of 17th March 1996 (Sunday), 30 man working days are finished.

Both of them work on Monday,

S works on Tuesday

Both of them work on Wednesday

S works on Thursday

Both of them work on Friday and the remaining 8 man working days are also over.

Hence, the required date is $17+5 = 22$ March 1996 (Friday)

[VIEW SOLUTION](#)



CAT Syllabus PDF

